Peptide collapse

Turning a Spectronaut precursor report into the wide phosphosite matrix the rest of the tool expects, without touching the session.

AT A GLANCE

Where	External Tools → Peptide Collapse. Not in the DataViewer, which keeps only **GONetView** in this section.
Takes	One Spectronaut report, tab-separated, chosen from disk. It is not one of the four uploaded datasets and it does not become one.
Gives	One tab-separated file, named after the input with _collapsed.tsv in place of its extension, offered as a download as soon as the run finishes.
Runs	Inside the tool, on the machine running it. Nothing is sent anywhere, and nothing is kept: the server's working copy is deleted when the download is served.
Changes	Nothing in the session. The collapsed file re-enters the analysis only when you upload it yourself, as the phospho level.
One setting	**Localization cutoff**, 0 by default. 0.75 is the Class I convention.

1 What it is for

Spectronaut writes one row per precursor per run, and a phosphosite is not a precursor: the same site is measured through several charge states, several missed cleavages and several co-modified forms, each of them a separate row. Every phospho page in this tool reads a wide matrix whose rows are sites, so something has to decide which precursors are the same site and what one number for that site in one run is. That is this page, and the reason it sits under **External Tools** rather than under **Data** is that it works on a file rather than on the loaded analysis: it reads nothing from the session and writes nothing to it.

The collapse is therefore a step *before* the tool, run from inside it. The output is an ordinary file on your disk, and it becomes data only at Data → Upload.

> **KEEP THE TWO STEPS SEPARATE**
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> Choose the precursor report, set the localisation cutoff, and run the collapse. Download the resulting site table and inspect its identifiers and sample columns. Upload that table as phosphosite data to begin the analysis; finishing the collapse does not itself replace a loaded dataset.

> **NOTE**
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> This is the collapse that produces phosphosites from precursors. Collapsing phosphosites into phosphoproteins is a different operation on an already uploaded dataset, with its own four methods, and is at Data → Annotation under **Aggregate to Phosphoprotein**.

2 Running it

1. Choose the report under **Spectronaut TSV file**. The file dialogue is restricted to .tsv and .txt; the file is read as tab-separated whichever of the two it is. Its size is bounded by the same upload limit as a dataset, which whoever deployed the instance can raise.
2. Set **Localization cutoff**. It is a probability between 0 and 1; anything outside that range is clamped as you type and refused by the server if it arrives anyway.
3. Press **Run Peptide Collapse**. The button reads Running… until the file comes back.
4. The download starts by itself. The same file stays available on the page as a link until you choose another input.

Controls on this page

Control	What it does	Default	Where
Spectronaut TSV file	The report to collapse. Choosing a new one clears the previous result.	none	page
Localization cutoff	The minimum site localization probability a row must reach to survive. 0 keeps every localized site; 0.75 is the Class I convention.	0	page
Run Peptide Collapse	Runs the collapse and downloads the result. Disabled until a file is chosen.	—	page

3 How it is computed

THE COLUMNS IT REQUIRES

Seven columns must be present, by these exact names: R.FileName, EG.PrecursorId, EG.TotalQuantity (Settings), PEP.PeptidePosition, EG.PTMAssayProbability, PG.Genes and PG.ProteinGroups. Each one is reported as present or absent in turn, and the first absent one stops the run by name. No column is guessed from a synonym here — this is the search engine's own report, not the tool's canonical naming.

FROM A PRECURSOR TO A SITE

Rows whose EG.PrecursorId carries no [Phospho (STY)] marker are dropped, and their count is reported. The precursor identifier is then reduced to a bare sequence: the charge suffix, every bracketed annotation and every underscore are removed.
A precursor carrying *n* phospho markers becomes *n* rows, one per modified residue. The position of each within the peptide follows from the lengths of the pieces between the markers, and the residue at that position — S, T or Y — is read off the bare sequence. The peptide's own start in the protein comes from PEP.PeptidePosition; where that field lists several positions, separated by semicolons or commas, the first is used. The site position in the protein is then

$$\text{site} = \text{position in peptide} + \text{peptide start} - 1.$$

Multiplicity is the number of phospho markers on the precursor, capped at 3: a precursor with four sites is counted as M3.

THE COLLAPSE KEY

Every derived row gets the key that defines the site:

PG.ProteinGroups~PG.Genes_<residue><position>_M<multiplicity>

which for the first site of a singly phosphorylated peptide reads P00001~GENEA_S123_M1. Rows sharing a key are the same site measured through different precursors, and they are what is collapsed.
The underscore is a separator of that key, so a gene name containing one would split it. Such names are rewritten with # before the key is built and rewritten back to _ afterwards, in both the key and the gene column; the count of affected names is reported.

THE QUANTITY

EG.TotalQuantity (Settings) is pivoted to one column per run, taking the first value for each precursor-and-position. Every column is then coerced to a number, so a cell Spectronaut wrote as Filtered becomes NA rather than stopping the run; a cell written as 0 is replaced by NA as well, because it is a sentinel for a measurement that was not made. Only then are the values taken to $\log_2$.

The order matters in both directions. Taking $\log_2$ first would turn a sentinel 0 into $-\infty$, which must not be treated as an observation; deleting zeros *after* the transform would delete a genuine intensity of 1, whose $\log_2$ is 0.

The collapsed value is then the *median of the* $\log_2$ *values* of the precursors sharing a key, per run — not the $\log_2$ of their median. A run in which no precursor of that site was measured keeps its column and holds NA; a run in which nothing at all was measured still appears, as an all-NA column.

LOCALIZATION AND THE CUTOFF

EG.PTMAssayProbability is reduced twice: first to the maximum across runs, then to the maximum across the precursors sharing a key. The site's reported PTM_localization is therefore the best evidence any run produced for it, not an average and not the value in the run being read.

Rows are kept where that value is at least **Localization cutoff**. A row whose probability is missing everywhere fails the comparison and is dropped, at every cutoff including 0.

Edge cases

- A row is discarded before collapsing if its peptide position is absent, its gene is empty, its protein group is empty, or the residue at the modified position could not be read. The count is reported as phosphopeptides filtered for insufficient information.
- Sample columns are named after R.FileName and are ordered as the runs first appear in the report. Renaming samples is the annotation's work, later.
- The page always collapses to protein groups. The function underneath can also collapse to individual proteins, splitting a group into one row per accession, but nothing in the interface selects that.
- Nothing is written back: the input copy and the result are made in a temporary directory that is removed once the download has been served, and both are gone if the run fails.

That is intended: a site with no localization probability anywhere has nothing to be confident about, so it is dropped at any cutoff, including zero.

Collapsing per accession rather than per protein group is not a supported route and the page does not offer it.

4 The file it produces

Columns in the collapsed file

Column	What it holds
PTM_Collapse_key	The site: protein group, gene, residue, position and multiplicity, as built above. This is the feature identifier every phospho page uses.
Protein_group	The first accession of the protein group.
Gene_group	The first gene name of the group, with any underscore restored.
UPD_seq	The bare peptide sequence with the modified residue lower-cased and marked with *, so the site can be read in sequence context.
PTM_localization	The highest localization probability observed for that site in any run.
One column per run	The collapsed $\log_2$ intensity of that site in that run, named after R.FileName. NA where nothing was measured.

> **WORTH KNOWING**
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> Those five annotation columns are exactly the five that **Aggregate to Phosphoprotein** requires before it will collapse phosphosites into phosphoproteins. A collapsed file uploaded unchanged carries them; a file that has been edited in a spreadsheet in between often does not, and the collapse then refuses it by name.

5 Getting the result into the analysis

1. Open Data → Upload.
2. Choose Phospho as the dataset type and the _collapsed.tsv file.
3. Upload, then annotate as usual: the run columns are the samples, and the five annotation columns above are not.

The values are already $\log_2$, so tick **Dataset is already log2 transformed** when the annotation is generated. The detector usually reaches the same conclusion from the value range, but the file states it and you know it.

6 Exporting

There is nothing to export afterwards — the run *is* the export. The file is tab-separated and written the same way every time, so two collapses of the same report can be compared line for line. The report has to be uploaded; there is no route that collapses a file named by its path on the server. In the DataViewer the whole page is absent.

Behind this page

Function	What it produces
peptideCollapse	Precursor rows to one $\log_2$ row per phosphosite, with the collapse key, the localization maximum and the cutoff
process_and_export	Reads the tab-separated report, collapses it and writes the result beside it
run_peptide_collapse	The page's call: resolves the output name, runs the collapse and reports rows, columns and the log